

PROJECT REPORT

FILE CONVERTER TOOL

A Browser-Based Image Format Conversion Tool

Submitted By	Manvi
Registration Number	26BHI10076
College	VIT-BPL
Submission Date	September 29, 2026

Submitted as a project report for academic evaluation.

1. ABSTRACT

The File Converter Tool is a simple browser-based application designed to convert images between JPG, PNG, and WEBP formats. The project is implemented using HTML5, CSS3, and JavaScript with the Canvas API. According to the supplied project documentation, the tool works entirely in the browser and does not require a backend server. It also supports drag-and-drop image input and is designed to be mobile friendly.

The supplied test program also demonstrates conversion-related functions, validation and error handling, an object-oriented FileConverter class, and a statistics/array structure. The recorded test run shows all 9 tests passing successfully.

2. INTRODUCTION

Images are commonly stored and shared in different file formats. A lightweight converter can make it easier to change an image into a format suitable for a particular use. This project provides a local browser-based approach so that image conversion can be performed without sending the selected image to a remote server.

The project is organized around a simple workflow: open the web page, upload an image, select the required target format, and convert/download the result.

3. OBJECTIVES

The main objectives of the project are:

- To convert images into JPG, PNG, and WEBP formats.
- To provide a browser-based conversion process without a backend.
- To support drag-and-drop image input.
- To provide a simple and mobile-friendly user experience.
- To implement conversion, validation, error handling, and testing logic.
- To verify the main functionality through automated-style project tests.

4. KEY FEATURES

The project documentation lists the following features:

- Image conversion to JPG, PNG, and WEBP.
- 100% browser-based operation; no data is uploaded to a server.
- Drag & Drop support.
- Fast and secure operation as described in the project README.
- Mobile-friendly design.

5. TECHNOLOGY STACK

Technology	Role
HTML5	Web page structure and interface.
CSS3	Presentation and responsive styling.
JavaScript	Application logic and conversion workflow.
Canvas API	Browser-side image processing/conversion support.

6. WORKING / USER WORKFLOW

The supplied README specifies the following usage sequence:

Step 1: Open index.html in a web browser.

Step 2: Upload the required image.

Step 3: Select the target image format.

Step 4: Click Convert & Download.

7. PROJECT STRUCTURE AND IMPLEMENTATION

The supplied project files show a converter module and a separate test program. The test program imports conversion functions including TXT/CSV/JSON conversion helpers, a FileConverter class, validation logic, and a statistics function. The testing file checks individual conversion operations as well as class-based file conversion and validation.

For the browser-facing project described in the README, the primary interface is index.html, while HTML5, CSS3, JavaScript, and the Canvas API form the stated technology stack.

8. TESTING AND VALIDATION

The supplied terminal screenshots show a successful test run for the converter project. The displayed test suite contains nine tests, and each test is marked PASSED.

Test	Area Checked	Result
1	TXT to CSV	PASSED
2	CSV to TXT	PASSED
3	TXT to JSON	PASSED

Test	Area Checked	Result
4	JSON to TXT	PASSED
5	CSV to JSON	PASSED
6	JSON to CSV	PASSED
7	FileConverter Class & OOP	PASSED
8	Error Handling & Validation	PASSED
9	Statistics & Array Structure	PASSED

Overall test result: ALL 9 TESTS PASSED SUCCESSFULLY (100% OK), as shown in the supplied terminal output.

9. ERROR HANDLING AND VALIDATION

The supplied test file explicitly checks invalid and incomplete inputs. The checks include an empty filename, empty content, an invalid conversion choice, a mismatched file extension, and invalid JSON syntax. These cases are expected to fail validation or return an error, and the corresponding tests are reported as passed.

10. OBJECT-ORIENTED COMPONENT

The test program includes a FileConverter class and checks its conversion workflow, output filename, success status, summary information, and saving of the converted file. This provides an object-oriented structure for handling a file conversion operation.

11. STATISTICS / DATA STRUCTURE CHECK

The supplied tests also include a statistics function that processes text and checks the number of lines, number of characters, and average line length. This demonstrates the use of basic array/list-style data handling and numerical calculation in the project test code.

12. RESULTS

The project documentation describes the browser tool as functional for image conversion to JPG, PNG, and WEBP. The supplied testing screenshots additionally show a successful 9-test verification run, with every displayed test marked PASSED.

13. ADVANTAGES

- Browser-based operation with no backend required.
- The README states that image data is not uploaded to a server.
- Simple four-step user workflow.
- Supports multiple common image formats.
- Drag-and-drop support improves ease of use.

- Mobile-friendly design is listed as a project feature.

14. LIMITATIONS

The supplied documentation is concise and does not specify advanced image-editing features, quality controls, batch conversion, or detailed browser compatibility information. These items are therefore not claimed as part of the current project.

15. SUGGESTED FUTURE SCOPE

The following are possible extensions that could be considered in a future version. They are suggestions rather than features documented as already implemented:

- Batch conversion of multiple images.
- Optional image-quality controls before download.
- Image resizing and basic compression options.
- More detailed conversion progress and user feedback.
- Additional output formats if required.

16. CONCLUSION

The File Converter Tool is a compact browser-based project for converting images into JPG, PNG, and WEBP formats. Its documented approach uses HTML5, CSS3, JavaScript, and the Canvas API without requiring a backend. The supplied test evidence further demonstrates successful execution of nine verification tests covering conversion, object-oriented functionality, validation/error handling, and statistics/data structures.

Overall, the project presents a clear and practical example of a client-side web utility with a straightforward user workflow and a dedicated testing component.

17. SOURCE FILES PROVIDED FOR THIS REPORT

This report has been prepared from the supplied project README, terminal test screenshots, and source-code screenshots provided with the project submission.

— END OF REPORT —